

Tanish Baranwal

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EDUCATION

University of Oxford

Oct 2026 – Sep 2030 (expected)

Incoming DPhil in Engineering Science

St Anne's College

- BOLD Lab · Supervisors: Profs. Jakob Foerster and João F. Henriques · Meta AI PhD Program

University of California, Berkeley

Aug 2022 – May 2025

Bachelor of Science in Electrical Engineering and Computer Sciences

Cumulative GPA: 3.98/4.0

- **Relevant Coursework:** Probability and Random Processes, Deep Neural Networks, Computer Vision, Operating Systems and System Programming, Physics-Inspired Machine Learning, Algorithmic Aspects of Machine Learning
- President, Eta Kappa Nu (EECS Honor Society), Spring 2025.

EXPERIENCE

GSA Capital

Jun 2026 – Sep 2026

Quantitative Research Intern

London, UK

- Built a distributed PyTorch training stack and high-throughput data pipelines for financial transformers; trained **16M–4B parameter** models and fitted and validated model/data scaling laws through controlled sweeps.
- Developed a multi-period portfolio optimizer incorporating market impact and model predictive control; derived a custom ADMM solver achieving **5–30× faster** rebalance loops than MOSEK at matched solution tolerances.
- Studied cross-sectional generalization and data pooling for nonlinear forecasting models; researched and developed new features incorporated into a production trading strategy.

Berkeley Artificial Intelligence Research

Sep 2024 – May 2026

Researcher; advised by Prof. Jitendra Malik

Berkeley, CA

- Led **Video-GMAE**, a self-supervised video model in which point tracking emerges from predicting 3-D Gaussians over time; **first author, CVPR 2026 Highlight**.
- Developed the architecture, training objective pipeline, evaluation framework; achieved **60.1 average Jaccard** on TAP-Vid Kinetics in zero-shot tracking without tracking labels, optical flow supervision, or bounding boxes.
- Explored natural-policy-gradient extensions using second-order methods; evaluated variants on MuJoCo tasks.

GSA Capital

Jun 2025 – Sep 2025

Quantitative Research Intern

London, UK

- Built a JAX execution model using generalized linear models and market signals to predict price improvement in broker smart-order routing and inform algorithm selection, reducing realized slippage by **15%**.
- Built an intraday trading strategy from scratch, covering alpha research, risk modeling, and portfolio optimization; validated in paper trading and deployed live by my portfolio manager shortly after the internship.

Amazon

May 2024 – Aug 2024

Software Development Engineer Intern

Seattle, WA

- Built and evaluated a distributed LLM inference system using **Amazon Bedrock** to infer and normalize missing footwear-size attributes across **100K+ product listings**, producing validated outputs for **85% of listings**.
- Architected an AWS inference and evaluation pipeline using Lambda, S3, Apache Spark, and EMR; incorporated batching and model selection to support a target scale of **100M+ catalog items**.

Cardiac Vision Labs, University of California, San Francisco

Sep 2022 – Jun 2025

Computer Vision Researcher

San Francisco, CA

- Developed diffusion models for cardiac excitation-wave dynamics, supporting conditional generation, temporal evolution, inpainting, and 3-D reconstruction from partial observations.
- First-authored research in **APL Machine Learning**; presented at Computing in Cardiology 2023 and the SIAM Annual Meeting 2024.

SELECTED PUBLICATIONS

Baranwal, T., Singh, H. G., Rajasegaran, J., & Malik, J. (2026). Tracking by Predicting 3-D Gaussians Over Time. *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. **Highlight**.

Baranwal, T., Lebert, J., & Christoph, J. (2024). Dreaming of Electrical Waves: Generative Modeling of Cardiac Excitation Waves using Diffusion Models. *APL Machine Learning*.

SKILLS AND INTERESTS

Skills: Python, PyTorch, JAX, Distributed training, DeepSpeed, C++, TensorFlow, AWS, XGBoost, MATLAB

Interests: Photography, Coffee, Weightlifting, Basketball, Swimming, Mixology, Cooking